Federal Reserve Announcement Days and US Equity Volatility: An Event Study of FOMC Statements, 2016–2026

SIQuant Research[1]

[1]Hypothesis × Quant

October 26, 2026

Abstract

Hypothesis: H_0: mean |return| of the S&P 500 on FOMC announcement days equals that of ordinary sessions; H_1: it is larger

Data: 81 FOMC statements (88 listed) matched to S&P 500 and VIX closes, 2016-11-02 to 2026-09-16

Method: 10,000-draw placebo over ordinary sessions (meeting ±1 day excluded); Bonferroni $\alpha = 0.025$ for two outcomes

Results: Mean |return| 0.846% versus placebo 0.724% (one-sided $p = 0.1021$); mean ΔVIX 0.64% versus placebo 0.36% ($p = 0.6275$ for a larger fall).

Research decision: **REJECTED**

1 Design

FOMC statements are released at 14:00 US/Eastern, two hours before the cash close, so the close-to-close return of the release date contains the full announcement reaction together with 5.5 hours of pre-announcement trading, which dilutes the measured effect. The placebo draws ordinary sessions from the same sample, so volatility regimes are matched on average [Lucca and Moench, 2015].

2 Results

Table 1: S&P 500 and VIX on FOMC announcement days by policy action

Action	Events	Mean |return|	Mean return	Mean ΔVIX
HOLD	51	0.73%	0.03%	0.14%
HIKE	20	1.11%	0.01%	-1.73%
CUT	10	0.90%	-0.55%	7.91%

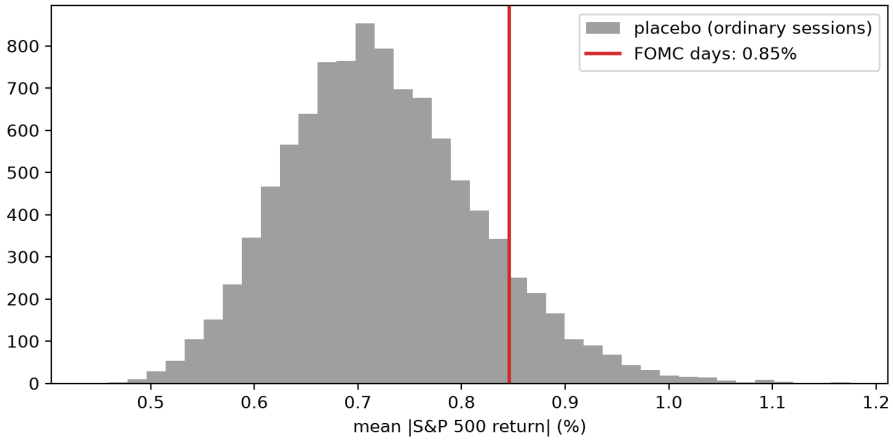


Figure 1: Mean absolute S&P 500 return on FOMC days against the placebo distribution.

The signed mean return on FOMC days is -0.048% (t=-0.36).

3 Limitations

The sample is short (about ten years, one tightening cycle) and clustered in 2020 when unscheduled meetings coincided with the COVID shock; a regime-matched placebo is left to follow-up work. Intraday data would isolate the announcement window but are not free.

References

David O Lucca and Emanuel Moench. The pre-fomc announcement drift. *The Journal of Finance*, 70(1):329–371, 2015.